

OG BONNAH CHINEMEREM

AUTOMATION & CONTROL ENGINEERING STUDENT
| PLC, DCS, ROBOTICS & INDUSTRIAL AUTOMATION |



MY CONTACT

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MY PROFILE

Third-year Automation and Robotics Engineering student at JAMK, building hands-on expertise across PLC programming, DCS systems, industrial robotics, and machine vision. Experienced with industry-standard tools, including Siemens TIA Portal, Beckhoff TwinCAT, Valmet DNA, and ABB RobotStudio. Driven by a genuine interest in designing reliable, efficient automation solutions — and comfortable working across the full project lifecycle from documentation and design through to commissioning and troubleshooting.



TECHNICAL SKILLS

Automation & Control Systems

- PLC Programming (Ladder Logic, Function Block Diagram)
- Siemens TIA Portal
- Beckhoff TwinCAT
- Valmet DNA (DCS Systems)
- PID Control
- Sequence Control
- HMI / SCADA Development
- OPC UA
- Aveva Intouch
- Siemens WIN CC

Engineering & Development Tools

- AutoCAD
- LabVIEW
- Git
- Visual Components

Robotics & Automation Systems

- ABB RobotStudio
- Visual Components
- MiR Programming
- Hazard and Risk Analysis
- Robot Simulation and Cell Design

Machine Vision & Image Processing

- Optical Character Recognition (OCR)
- Object Sizing and Dimensional Measurement
- Image Processing and Analysis
- Vision-Based Inspection Systems

Programming

- C#
- Python
- MATLAB

Industrial Systems & Instrumentation

- Sensors and Transmitters
- Control Valves
- Pumps and Actuators
- I/O Module Configuration
- Signal Scaling and Data Conversion
- Field Instrument Design
- Testing, commissioning and diagnostics.

Industrial Communication Networks

- PROFINET
- PROFIBUS
- Modbus TCP
- AS-i Bus
- EtherCat
- Network Configuration



EDUCATIONAL BACKGROUND

Bachelor's of Automation and Robotics Engineering
JAMK University of Applied Sciences

2023 - 2027

Deeper Life High School
High School

2016-2022



LANGUAGE

English -> Fluent

Finnish -> Intermediate



AUTOMATION PROJECTS

- **Water Process Automation Module – Valmet DNA**
 - Developed a process automation module using Valmet DNA
 - Implemented PID control loops, cascade control, and ratio control strategies
 - Designed process interlocks and safety logic
 - Created loop structures integrating pumps, transmitters, and control valves
 - Performed signal scaling, process data handling and implemented automated sequence control



AUTOMATION PROJECTS

- **PLC Controlled Pumping Module – Siemens TIA Portal**
 - Designed and programmed a PLC automation system using Siemens TIA Portal
 - Implemented control logic for valves, pumps, and sensors using device documentation and data sheets
 - Integrated VFD control and PID regulation
 - Developed HMI and SCADA visualisation using WinCC
 - Configured AS-i bus field communication
 - Implemented automated sequence control for process operation
 - Troubleshooting and fault correction
- **Process control using Aveva Intouch HMI and Siemens PLC through OPC UA (Ongoing)**
 - Creating a HMI application for a Siemens-controlled water process module using OPC UA to communicate between both devices.
 - Using faceplates and user-defined types to improve the reusability of the program
 - Utilised AI to build an automated tool to identify and convert TIA portal tags to an Aveva usable format
- **Automation Engineering Documentation**
 - Field devices planning and documentation, including creating I/O lists, wiring and circuit diagrams, cross-connection tables, field cabinet layouts and necessary modifications and accessories.
- **Robotics & Automation Systems**
 - Programmed and simulated industrial robotic systems using ABB RobotStudio
 - HAZOP Analysis and risk reduction for a robot cell environment
 - Developed robotic production simulations using Visual Components
 - Worked with MiR mobile robot systems and UR Cobots
- **Machine Vision Systems**
 - OCR recognition for automated text reading
 - Object dimensioning and measurement for quality control
 - Edge detection and pattern recognition for quality inspection
- Built and configured network structures in a virtual environment, including routing, switching, firewalls, and IPv4/IPv6 addressing.



WORK EXPERIENCES

Talent Boost Intern - JAMK

Aug 2025

Supported event planning and developed initiatives to improve student adaptation to the Finnish working environment. Collaborated with companies, departments, and external organizations to organize and host events, managing communications, meetings, and stakeholder coordination.

Marketing Intern - UMT Victor

Nov 2024 - Mar 2025

Supported marketing operations within a Finnish business environment, gaining experience in professional communication and workplace practices.

Viitasaari Hackathon

May & Sept 2025

Worked in a team to develop technical solutions for real company cases, presented our progress to the company at regular intervals, and gathered feedback to keep the solution on track.

Media Club - Deeper Life High School

2019 - 2022

Assisted the media team with setting up equipment for events and solving technical issues during the event, using temporary fixes when needed until proper repairs could be made.



REFERENCES

Anna Tenhunen
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